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Research paper



Development of an early stage hydrofoil design tool including unsteady hydrodynamics

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ABSTRACT

Conceptual design of hydrofoil craft and in particular hydrofoils is challenging since the hydrodynamics is governed by multiple complex physical phenomena such as the effect of the free surface, turbulence and unsteadiness due to waves and/or operations. High-fidelity simulations capturing these physical phenomena are computationally too demanding and costly to be used directly in conceptual hydrofoil design. An early stage hydrofoil design tool is developed by training a recurrent neural network on the unsteady lift and drag due to forced pitch motions and due to forced heave motions of a geometrically parameterized hydrofoil. By combining static and dynamic features, the machine learning model learns the unsteady response of the hydrofoil for an arbitrary set of parameters describing the geometry. The mean relative test error with respect to high-fidelity RANS results in lift and drag due to forced pitching is 13.16 % and 9.67 % respectively. For forced heave, we find 1.78 % and 5.78 % respectively. Evaluation of the machine learning models is very cheap. This makes it possible to solve multi-objective optimization problems in a limited amount of time by using the machine learning model as a surrogate in the optimization process. A web application is developed for this purpose, resulting in an useful tool for conceptual hydrofoil design. This development is followed by a verification and validation case and a case study to assess the value of the tool in practice. The tool can quickly propose new and interesting hydrofoil designs with favorable performances for given operational conditions. This motivates to explore new machine learning strategies and enhance tool development with the purpose to improve conceptual hydrofoil design.

1. Introduction

Lifting surfaces operating below the free surface are mainly used to control marine vehicles. A special type of lifting surface is the hydrofoil which is designed to support the weight of a ship thereby reducing ship resistance dramatically and increasing comfort due to reduced motions. The flow around such a submerged hydrofoil is highly influenced by the free surface. The influence of the free-surface on the foil's hydrodynamic performance strongly depends on the problem parameters such as Froude number, submergence depth to chord ratio and angle of attack, see Pernod et al. (2023), Perali et al. (2024), Nicolas et al. (2023) and Bot et al. (2025). Unsteadiness due to operational conditions (such as waves and/or motions) complicate the hydrodynamics even more. Unsteady, viscous and turbulent fluid flow around a plunging hydrofoil near the water free surface for a hydrofoil undergoing different submergence depths and oscillation frequencies was studied in Esmaeilifar et al. (2017). It was concluded that the foil drag may significantly increase in a range of critical unsteady parameters for which large unsteady waves are generated. On the other hand the foil may produce thrust for other unsteady parameters due to the Knoller–Betz or Katzmayr effect, see Jones et al. (1998).

While designing a hydrofoil these complex physical phenomena need to be accounted for using high-fidelity simulations which are often computationally demanding. The design of a hydrofoil therefore often starts with simplified and cheaper low-fidelity models before high-fidelity models are used. This strategy may lead to sub-optimal hydrofoil designs since important physical phenomena which are not captured by low-fidelity models may be important already since the early conceptual design stage. To be able to include high-fidelity information earlier in the design process Simulation Based Design Optimization (SBDO) techniques can be used replacing the high-fidelity model with a cheap surrogate, see Serani et al. (2024) and Ploé (2018). Sometimes the low-fidelity and high-fidelity model are exploited together in the construction of a so called multi-fidelity surrogate model, see for example Perali et al. (2024), Solak et al. (2024) and Wackers et al. (2023). Another popular SBDO technique relies on direct gradient

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Nomenclature

α	Pitch (rad)
ξ_0	Space fixed position vector (m)
λ_{cr}	Camber ratio (–)
λ_s	Submergence ratio (–)
λ_{tap}	Tapering ratio (–)
λ_t	Thickness ratio (–)
λ_{y_k}	Kink distance ratio (–)
μ	Viscosity of seawater ($kg\,m^{-1}\,s^{-1}$)
Φ	Disturbance potential ($m^2\,s^{-1}$)
Φ_T	Velocity potential ($m^2\,s^{-1}$)
Φ_w	Incident-wave potential ($m^2\,s^{-1}$)
ρ	Density of seawater ($kg\,m^{-3}$)
σ_{bend}	Maximum normal stress due to bending ($kg\,m^{-1}\,s^{-2}$)
σ_{bucket}	Cavitation threshold (–)
$\sigma_{N,neutral}$	Cavitation number at neutral angle of attack (–)
σ_{risk}	Cavitation risk (–)
σ_{yield}	Yield stress ($kg\,m^{-1}\,s^{-2}$)
$\sigma_{N,pressure}$	Cavitation number on pressure side (–)
$\sigma_{N,suction}$	Cavitation number on suction side (–)
A	Cross sectional area (m^2)
A_h	Peak-to-peak heave amplitude (m)
a_h	Heave amplitude (m)
A_r	Aspect ratio (–)
a_α	Pitch amplitude (rad)
c	Mean aerodynamic chord (m)
c_l	Local chord (m)
c_{root}	Root chord (m)
c_{tip}	Tip chord (m)
C_D	Drag coefficient (–)
C_L	Lift coefficient (–)
d	Submergence distance (m)
F	Force component acting on hydrofoil ($kg\,m\,s^{-2}$)
f_α	Pitch frequency (s^{-1})
f_h	Heave frequency (s^{-1})
F_n	Froude number (–)
g	Gravitational acceleration ($m\,s^{-2}$)
h	Heave (m)
h'	Non-dimensional heave (–)
I	Area moment of inertia (m^4)
$k_{h/\alpha}$	Reduced frequency (–)
M_{max}	Maximum bending moment on the foil ($kg\,m^2\,s^{-2}$)
p	Pressure ($kg\,m^{-1}\,s^{-2}$)
Re	Reynolds number (–)
S	Projected planform area (m^2)
S_t	Strouhal number (–)
t'	Non-dimensional time (–)
t_{max}	Maximum thickness of the foil profile (m)
U_∞	Free stream speed ($m\,s^{-1}$)
y_k	Kink distance (m)
Z_u, Z_l	Upper and lower foil profile (m)
cam_{max}	Maximum camber of the foil profile (m)

based optimization with the gradients computed from adjoints, see for example Garg et al. (2015) and Liao et al. (2022).

A number of early stage design tools can be found in the literature. Typhoon (Bagué et al., 2021) is an open-source preliminary design tool implementing the vortex-lattice method and is mainly useful for dynamic stability analysis. A machine learning enabled foil design assistant was proposed in Kostas and Manousaridou (2023), implementing a parametric model with 17 parameters to describe a variety of profile shapes. A machine learning model was trained mapping the input vectors to performance objectives obtained from the well known XFOIL computational package. Subsequently, the model was used for performance-based design optimization. It was shown that significant time savings can be obtained by using these machine learning models in the early design stage. In Li et al. (2024) and Wang et al. (2022), deep learning is used to build a tool for tidal turbine rotor design. In this work, Generative Adversarial Networks (GANs) are used to generate new designs whereas deep convolutional neural networks are used to learn the mapping from design to performance objectives.

In this contribution we aim to develop an efficient hydrofoil design tool able to predict the unsteady loads of a parameterized hydrofoil undergoing uncoupled heave and pitch oscillations. To this aim, we will use multi-fidelity recurrent neural networks combining the low-fidelity output of cheap unsteady vortex lattice simulations with the unsteady output of expensive high-fidelity RANS simulations. The focus on recurrent neural networks is motivated by the demonstrated potential of these methods when used for time series prediction in various engineering applications, see Marlantes et al. (2024), Li et al. (2021) and Liu et al. (2024). The focus on a multi-fidelity approach originates from the desire to use computational resources efficiently. Multi-fidelity recurrent neural networks are simple and elegant solutions to achieve this, see Nasser et al. (2025).

The resulting high-fidelity conceptual design tool accounts for viscous flow effects while consuming negligible computational resources. This will help the hydrofoil designer with early shape optimizations and control designs. In Section 2 an overview is given on the main components of the simulation based design optimization tool. Each component is further detailed in Sections 2.1, 2.2, 2.3 and 2.4 describing shape parametrization, numerical simulations, surrogate models and optimization respectively. In Section 3 the conceptual tool development is described followed by Section 4 on tool verification and validation. A case study is presented in Section 5, demonstrating the usefulness of the conceptual design tool in practice. Main conclusions are formulated in the last section.

2. Simulation based design optimization of a hydrofoil

The conceptual design tool should be able to rapidly repeat SBDO cycles for varying optimization objectives and constraints. Moreover, the tool should be able to explore diverse designs for a large variety of applications such as the design of foils for fast passenger ferries, energy-efficient workboats or sailing yachts. The main ingredients to build such a tool are identified as:

1. **Shape parametrization:** The parametrization should be rich enough to cover a wide variety of designs, but the size of the design space should be limited (preferably $N \leq 10$) in order to keep the computational effort of optimizing the surrogate small. An optimization with the surrogate should not take longer than 10 min in order to prevent large turn-around times when the SBDO cycle is repeated during the design of the hydrofoil.
2. **Numerical solver:** The numerical solver should be able to capture the physics at high fidelity for designs resulting from the shape parametrization, but the computational cost is restricted by the available budget for the creation of the tool. A maximum number of 200 high-fidelity RANS simulations were used in this contribution.

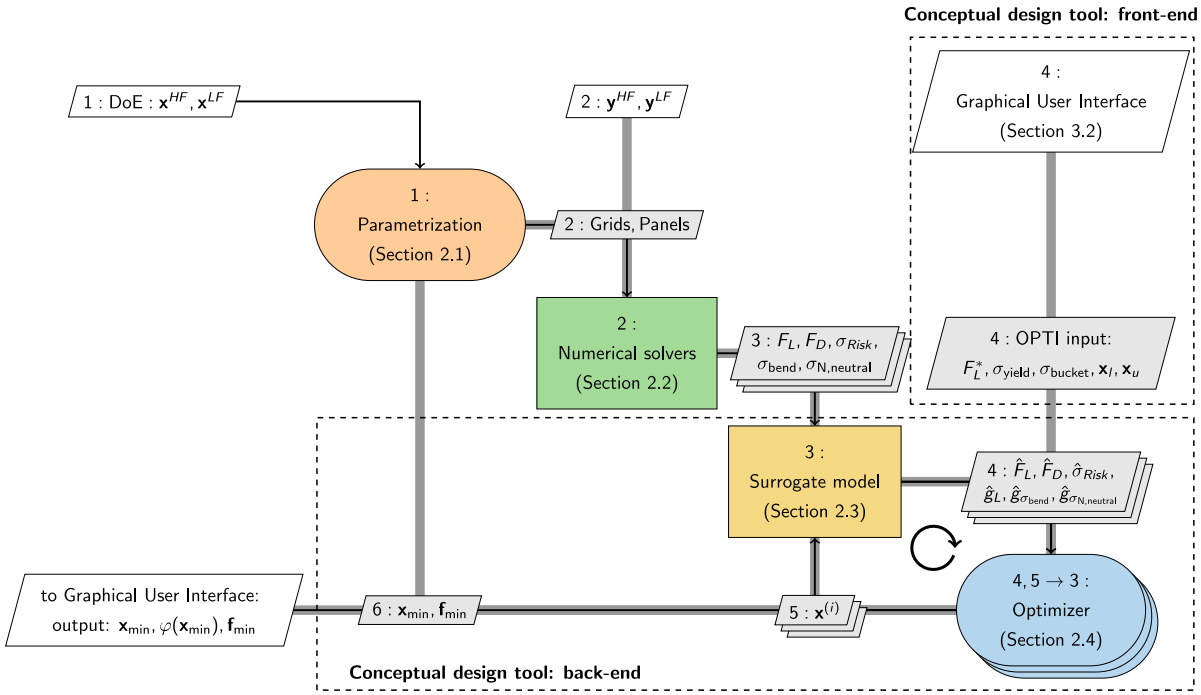


Fig. 1. XDSM diagram of the SBDO process leading to the conceptual design tool. The related sections of the paper are mentioned in brackets.

3. **Surrogate model:** Using supervised learning, the surrogate model is trained using numerical simulation data. The surrogate should predict the unsteady performance accurately for a given shape parameter, but the computational effort to evaluate the model should be a fraction of the cost of a high-fidelity simulation in order to keep the duration of the optimization below 10 min.
4. **Optimizer:** The optimizer evaluates the surrogate in an iterative manner, resulting in a generation of optimal and feasible designs. This process is repeated within the tool and should be very quick. The number of surrogate evaluations used by the optimizer should therefore be limited to keep the optimization time below 10 min.

With a restricted optimization time of 10 min, a trade-off should be made between the size of the design space, the cost of evaluating the surrogate and the number of allowed evaluations during the optimization. Fig. 1 illustrates the SBDO process using an Extended Design Structure Matrix (XDSM) to show data dependency and process flow on a single diagram, see Lambe and Martins (2012). The simulations (in component 2) are only evaluated once because the results are use to train a surrogate model (in component 3) which replaces the expensive simulations during optimization. Components 1, 3 and 4 are repeatedly evaluated for multiple optimization problems defined by the user in the graphical front-end of the design tool. These optimization problems are solved in the back-end of the conceptual design tool and the results are forwarded to the GUI for visualization. The following subsections explain the functionality of the components in Fig. 1, where the numbers in the components correspond to the subsection numbers.

2.1. Shape parametrization

Foils are parameterized using Grasshopper which is a graphical algorithm editor available in Rhinoceros 3D, see McNeel et al. (2023). Six geometrical parameters uniquely define each foil geometry. The foil definition begins drawing the 2D profile using the local chord (c_l), the maximum camber ($cam_{\max}$) and maximum thickness ($t_{\max}$). Fig. 2 presents a schematic representation of a generalized foil profile,

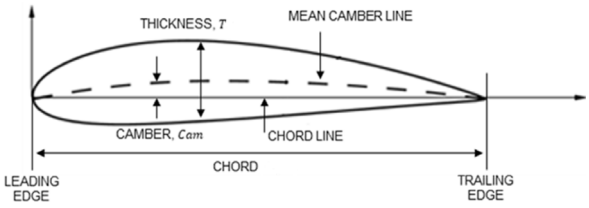


Fig. 2. Schematic Foil profile.

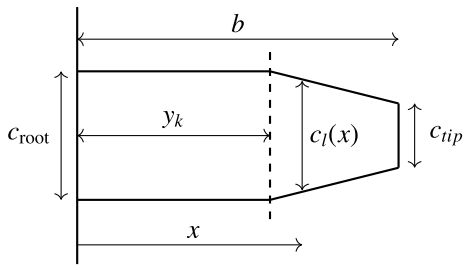
including the key dimensional parameters. In actuality, the thickness and camber curves follow the Eppler 817 profile. The predictions of the tool are therefore bounded to the wing profiles similar to Eppler 817 (or its scaled versions). Prediction performance of hydrofoil with different profile might differ.

Fig. 3 shows the simplified planform geometry and a side view of the submerged hydrofoil. The local chord c_l of the foil varies between the root chord and the tip chord, that is $c_l \in [c_{\text{tip}}, c_{\text{root}}]$ as can be seen in Fig. 3(a). The distance from the root to the location where the tapering of the hydrofoil starts, defines the kink distance y_k whereas the distance from root to tip defines the hydrofoil span b. With the projected planform area denoted by S, the mean aerodynamic chord of the hydrofoil wing is approximated by

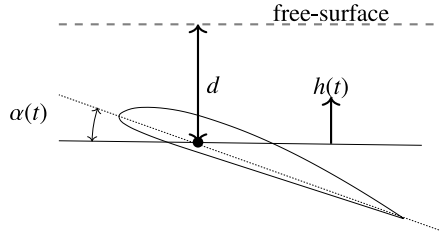
$$c \approx \frac{S}{b}. \tag{1}$$

Knowing the kink distance, wing span and planform area, two chord lengths are sufficient to uniquely define the wing geometry. Hence, from the chord lengths c, c_{tip}, c_{root} one parameter can be ignored in the feature set. In the following we use c and c_{tip} as chord features. This results in the tapering ratio c_{tip}/c as a predefined non-dimensional feature.

Fig. 3(b) shows the hydrofoil from the side where the submergence distance d is the distance from the free surface to the rotation point placed at 1/4 of the root chord. The dynamic pitch $\alpha(t)$ and dynamic heave $h(t)$ are prescribed rotations and translations of the hydrofoil with respect to this point.



(a) Planform geometry



(b) Side view of the submerged foil

Fig. 3. Hydrofoil planform geometry and side view.

2.1.1. Dimensional analysis

The force on the moving hydrofoil is written as a function of geometrical and operational input features, such that

$$F = f\big(\underbrace{\text{cam}_{\max}, c, c_{\text{tip}}, t_{\max}, y_k, S, b}_{\text{geometrical }(\mathbf{x})},\ \underbrace{g, \rho, \mu, U_\infty, d, h(t), \alpha(t)}_{\text{operational }(\mathbf{y})}\big),$$

where F is a force component acting on the foil, e.g. lift or drag. The operational features are given by the gravitational constant g, the density of water ρ, the viscosity of seawater μ, the free stream velocity U_∞, the submergence distance d and the dynamical input features consisting of the dynamic heave or the dynamic pitch given by

$$h(t) := a_h \cos(2\pi f_h t) \quad \text{or} \quad \alpha(t) := a_\alpha \sin(2\pi f_\alpha t), \tag{2}$$

where a_h and a_α denote amplitudes and f_{h} and f_α frequencies of the heave and pitch motion respectively.

Dimensional analysis is performed using the *BuckinghamPy* tool, see Karam and Saad (2021). In total, 14 dimensional parameters are identified. The fundamental dimensions of these parameters are Mass, Length and Time. By subtracting the number of fundamental dimensions from the total number of dimensional parameters it is found that 11 non-dimensional parameters describe the problem uniquely. Four non-dimensional length parameters are predefined. These are:

$$\lambda_{\text{tap}} := \frac{c_{\text{tip}}}{c}, \quad \lambda_{\text{t}} := \frac{t_{\max}}{c}, \quad \lambda_{y_k} := \frac{y_k}{b} = \frac{y_k c}{S}, \quad \lambda_{cr} := \frac{\text{cam}_{\max}}{c},$$

denoting the tapering ratio, the thickness ratio, kink distance ratio and the camber ratio respectively. The remaining 7 parameters are found with Buckingham-Pi:

$$\pi_1 := \frac{cg}{U_\infty^2}, \quad \pi_2 := \frac{\mu}{U_\infty c \rho}, \quad \pi_3 := \frac{F}{U_\infty^2 c^2 \rho},$$
$$\pi_4 := \frac{S}{c^2}, \quad \pi_5 := \frac{d}{c}, \quad \pi_6 := \frac{A_h}{c}, \quad \pi_7 := \frac{c f_h}{U_\infty},$$

where the peak-to-peak heave amplitude is given by

$$A_h := 2a_h. \tag{3}$$

The list is not unique, since 66 different sets of dimensionless parameters are found with the BuckinghamPy tool. A set is chosen such that the parameters are close to conventional non-dimensional parameters. This results in the following remaining non-dimensional parameters:

$$F_n := \frac{U_\infty}{\sqrt{cg}}, \quad Re := \frac{\rho U_\infty c}{\mu}, \quad C := \frac{F}{0.5\rho U_\infty^2 S},$$

$$A_{\text{r}} := \frac{S}{c^2}, \lambda_{\text{s}} := \frac{d}{c}, \quad S_{\text{t}} := \frac{f_h A_h}{U_\infty}, \quad k_{h/\alpha} := \frac{\pi f_{h/\alpha} c}{U_\infty},$$

denoting the Froude number, Reynolds number, force coefficient (lift and drag), aspect ratio, submergence ratio, Strouhal number and reduced frequency (for heave and pitch). In case of forced heave motions of the hydrofoil, the time dependent lift and drag coefficients are written as functions of non-dimensional geometrical parameters $\mathbf{x}'$ and operational parameters $\mathbf{y}'$

$$\begin{aligned} C_L(t') &= \mathcal{F}_L\big(\ A_r, \lambda_{y_k}, \lambda_{\text{tap}}, \lambda_{\text{cr}}, \lambda_{\text{t}},\ Fn, Re, \lambda_{\text{s}}, k_h, h'(t')\ \big) \\ C_D(t') &= \mathcal{F}_D\big(\underbrace{A_{\text{r}}, \lambda_{y_k}, \lambda_{\text{tap}}, \lambda_{\text{cr}}, \lambda_{\text{t}}}_{\text{geometrical}(\mathbf{x}')},\ \underbrace{Fn, Re, \lambda_{\text{s}}, k_h, h'(t')}_{\text{operational}(\mathbf{y}')}\big), \end{aligned} \tag{4}$$

where the non-dimensional heave and time are given by

$$h'(t') = \frac{S_{\text{t}}}{2}\cos(2k_h t') \quad \text{and} \quad t' = \frac{tU_\infty}{c}. \tag{5}$$

Similarly, in case of the forced pitch motions the time dependent lift and drag coefficient become

$$\begin{aligned} C_L(t') &= \mathcal{G}_L\big(\ A_{\text{r}}, \lambda_{y_k}, \lambda_{\text{tap}}, \lambda_{\text{cr}}, \lambda_{\text{t}},\ Fn, Re, \lambda_{\text{s}}, k_\alpha, \alpha(t')\ \big) \\ C_D(t') &= \mathcal{G}_D\big(\underbrace{A_r, \lambda_{y_k}, \lambda_{\text{tap}}, \lambda_{\text{cr}}, \lambda_{\text{t}}}_{\text{geometrical}(\mathbf{x}')},\ \underbrace{Fn, Re, \lambda_{\text{s}}, k_\alpha, \alpha(t')}_{\text{operational}(\mathbf{y}')}\big), \end{aligned} \tag{6}$$

with the non-dimensional pitch and time described by

$$\alpha(t') = a_\alpha \sin(2k_\alpha t') \quad \text{and} \quad t' = \frac{tU_\infty}{c}. \tag{7}$$

In this contribution we consider only uncoupled motions, so either forced heave motions or forced pitch motions. To learn the functions $\mathcal{F}_L(\mathbf{x}', \mathbf{y}')$, $\mathcal{F}_D(\mathbf{x}', \mathbf{y}')$, $\mathcal{G}_L(\mathbf{x}', \mathbf{y}')$ and $\mathcal{G}_D(\mathbf{x}', \mathbf{y}')$, low and high-fidelity numerical simulations need to be performed for various combinations of the non-dimensional geometrical and operational input features. This is the topic of the next section.

2.2. Numerical simulations

The development of computational methods for the calculation of hydrodynamic forces on hydrofoils goes back to the 1970's with the assumption of potential flow. These methods are based on a lifting line, on a Vortex Lattice (VLM) or a 3D lifting surface. A variety of methods exist with assumptions and simplifications that restrict their validity. Restrictions apply to for instance the inclusion of free surface effects and fore-aft foil interaction. Nowadays potential flow methods are capable to deal with such effects, see van Walree (1999). These fast computational methods can be extended to include viscosity effects based on empirical data. However, compared to full Computational Fluid Dynamics (CFD) simulations such as URANS, the accuracy of VLM is for example limited.

Details about the cheap but less accurate low-fidelity VLM simulations and the expensive but more accurate high-fidelity RANS simulations used in this study are found in Sections 2.2.2 and 2.2.3 respectively. We aim to create surrogate models exploiting both low-fidelity and high-fidelity simulations. These so called multi-fidelity methods pave the way to the development of efficient surrogate models, see Section 2.3. Crucial to the success of these methods is the use of a well designed plan of numerical experiments, this is the topic of Section 2.2.1.

2.2.1. Design of experiments

In this study, nested Latin Hypercube Sampling (LHS, see Qian (2009)) is used directly in the non-dimensional parameter space. The corresponding (non-unique) DoE for the physical variables is then constructed by assuming values for a few physical parameters and deriving the remaining physical parameters from the non-dimensional ones. More elegant ways of constructing the non-dimensional and

Table 1
Limits of the non-dimensional design-space.

Limit	Geometrical features ($\mathbf{x}'$)					Operational features ($\mathbf{y}'$)					
	Ar	λ_{y_k}	λ_{tap}	λ_{cr}	λ_t	Fn	Re	λ_s	$k_{h/\alpha}$	a_α	St_α
Lower	0.530	0.301	0.501	0.000	0.050	2.016	$1.01 \cdot 10^6$	1.008	0.011	0.021	0.000
Upper	14.982	0.799	0.800	0.050	0.300	13.988	$5.99 \cdot 10^6$	4.995	0.400	9.990	0.003

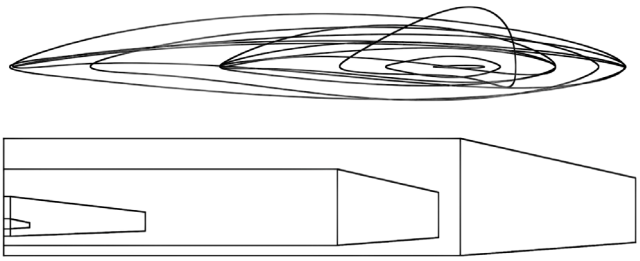


Fig. 4. Hydrofoil geometries $\phi(\mathbf{x}')$ for $\mathbf{x}' \in \mathbf{X}'$. Top: hydrofoil sections. Bottom: hydrofoil planforms. Each design has a corresponding operational feature vector $\mathbf{y}' \in \mathbf{Y}'$.

corresponding physical DoE simultaneously while accounting for domain constraints in both non-dimensional and physical space are found in Hazyuk et al. (2017). Such an approach is however beyond the scope of this paper.

A nested LHS design is defined to be a special LHS design that contains smaller LHS designs as subsets. The nested LHS approach introduces an incremental structure that is particularly useful when the optimal sample size for building a surrogate model is unknown. By progressively expanding the LHS sample size, we manage computational resources more effectively, addressing the high costs associated with URANS simulations. For simulations involving different levels of fidelity, the nested LHS guarantees overlapping design of experiments. This is a useful property when a multi-fidelity surrogate model needs to be learned from multi-fidelity simulation data.

A reduced set of non-dimensional features is defined prior to initiating the LHS (details in Section 2.1.1), ensuring that the data covers geometries and operational conditions encountered in practice. Hence, for each non-dimensional feature a typical range is chosen, see Table 1. For an impression of hydrofoil geometries $\phi(\mathbf{X}')$ that correspond to samples of a LHS design $(\mathbf{X}', \mathbf{Y}')$, see Fig. 4.

Starting with a smaller initial sample set and gradually increasing the sample size allows for careful monitoring of the model's performance. We define four tiers of sample size increments: 50, 100, 200, and 400. This tiered approach enables systematic scaling of the dataset while providing opportunities to evaluate model performance at each stage. After each increment, the model's accuracy and convergence behavior are assessed before deciding whether to proceed to the next level. For the training of multi-fidelity surrogates we choose the LHS designs consisting of 50 or 100 samples for the high-fidelity design of experiments $(\mathbf{X}', \mathbf{Y}')^{HF}$ whereas the LHS design consisting of 400 samples is used for the low-fidelity design of experiments $(\mathbf{X}', \mathbf{Y}')^{LF}$. Due to the nesting, we have $(\mathbf{X}', \mathbf{Y}')^{HF} \in (\mathbf{X}', \mathbf{Y}')^{LF}$, since the high-fidelity DoE is a subset of the low-fidelity DoE. Using these LHS designs, we strike a balance between generating a comprehensive dataset and managing the computational demand associated with CFD simulations.

2.2.2. Low-fidelity VLM simulations

Using VLM, the fluid motions are described by a velocity potential Φ_T:

$$\Phi_T(\xi_0, t) = \Phi(\xi_0, t) + \Phi_w(\xi_0, t), \tag{8}$$

where ξ_0 is the space-fixed position vector, Φ is the disturbance potential and Φ_w is the incident wave potential. The disturbance potential satisfies the Laplace equation and linearized boundary conditions are imposed on the free surface whereas tangential flow is imposed on the lifting surface. Impulsive strength vortex elements are used, allowing for a Green's function formulation, see van Walree (1999). This results in a boundary integral equation which, together with an appropriate wake model, has a unique solution for the unknown vortex strengths that can be obtained numerically. Once the vortex strengths are known, the forces acting on the lifting surface are computed by integrating the pressure p obtained from the unsteady flow Bernoulli equation:

$$\frac{p_{\text{ref}} - p}{\rho} = \frac{1}{2}(\nabla \Phi_T)^2 + \frac{\partial \Phi_T}{\partial t}. \tag{9}$$

To account for the effect of the foil thickness, an empirical correction is used. This reduces the accuracy of the method for foils having a large thickness to chord ratio. The unsteady solution is found using a sophisticated time-stepping process capable of introducing new wake vortex elements when gaps occur due to the motion of the lifting surface. The method requires far less computational resources than necessary for unsteady RANS but often results in significantly lower accuracy due to the thickness effect and the absence of viscosity. When the lift and drag on the hydrofoil are approximated using low-fidelity VLM computations, they are denoted by $C_L^{LF}(t)$ and $C_D^{LF}(t)$.

2.2.3. High-fidelity RANS simulations

RANS calculations are carried out using the viscous flow solver ReFRESCO, see Vaz et al. (2009) and Klaij and Vuik (2013). ReFRESCO is a CFD software package, developed at the Maritime Research Institute Netherlands (MARIN) in collaboration with several universities and partners. It solves unsteady (in)compressible viscous flows based on the Navier–Stokes equations, complemented with turbulence models and volume-fraction transport equations for different phases. In this contribution, simulations were carried out using the k-ω SST 2003 turbulence model. Pitch motions of the hydrofoil were handled using a grid deformation technique based on Radial Basis Functions, see de Boer et al. (2007) and Rendall and Allen (2008). Heave motions were applied by the moving grid technique which is a solid body movement of the entire mesh, including the surface mesh on the hydrofoil, all interior points and the mesh on the domain boundaries. The time step used in the unsteady simulations is obtained by dividing the period of the prescribed motion by 400 time steps.

Due to the symmetry of the foils, only half of the geometries are modeled. The computational domain extends $100 \cdot c_{\text{root}}$ in length and $32 \cdot c_{\text{root}}$ in width and height, see Peyro (2011). These dimensions are sufficiently large to avoid influence of the boundaries on the flow around the foil. An inflow and outflow boundary condition is placed on the inlet and outlet of the domain respectively. A symmetry boundary condition is placed on the symmetry plane and a constant pressure is defined for all the other boundaries.

To ensure grid similarity between geometries and consequently reduce the influence of discretization error, the cells size in the volume around the foil are set to $t_{\max}/10$. The free surface and other surfaces are then refined to ensure enough resolution while coarser grid is placed in the rest of the domain to reduce the number of cells. To ensure wall resolved flow, the y^+ is set lower than 1 on all surfaces. The number of grid cells in the entire domain depends strongly on the geometry of the hydrofoil (especially the thickness to chord ratio) and varies from 5 million to 40 million cells.

This method provides greater accuracy compared to VLM simulations at the costs of more effort in the set up and more computational

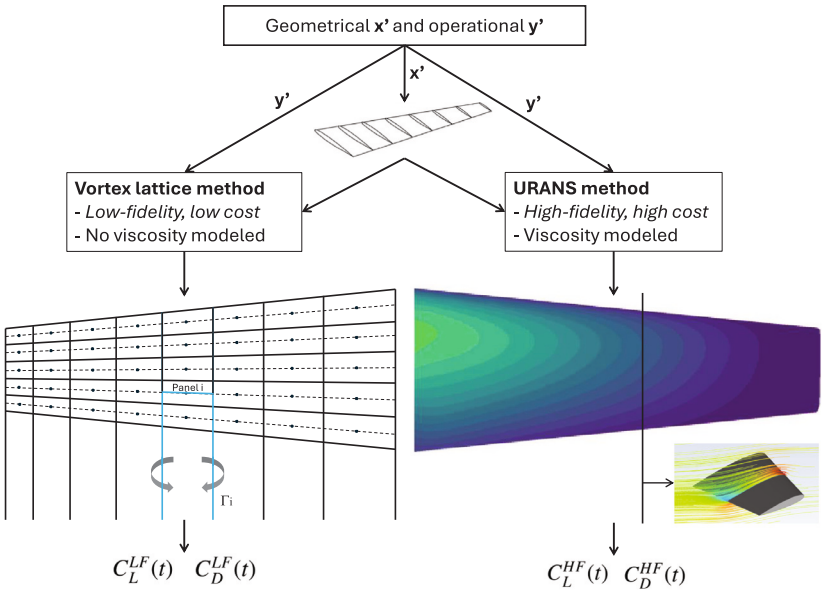


Fig. 5. Schematic of the simulation workflow. Left: the low-fidelity VLM method with the vortex strengths Γ_i that need be solved at the control points of the panels. Right: The high-fidelity URANS method. In contrast to the VLM solution, the URANS solution is obtained in a CFD domain around the foil, accounting for viscosity.

resources. Roughly speaking, one VLM simulation takes a few minutes between pre-processing and execution while a RANS simulation takes on average a couple of hours. When the lift and drag are approximated using high-fidelity RANS computations, they are denoted by $C_L^{HF}(t)$ and $C_D^{HF}(t)$. A schematic of the simulation workflow is given in Fig. 5, showing how the low and high-fidelity hydrodynamic coefficients are obtained from the geometrical and operational features defined in Section 2.1.1.

2.3. Surrogate models

The goal is to learn surrogate models approximating the functions $\mathcal{F}_L(\mathbf{x}',\mathbf{y}')$, $\mathcal{F}_D(\mathbf{x}',\mathbf{y}')$, $\mathcal{G}_L(\mathbf{x}',\mathbf{y}')$ and $\mathcal{G}_D(\mathbf{x}',\mathbf{y}')$ defined in Section 2.1.1 using the output of the numerical experiments defined in Section 2.2, while aiming for the lowest model prediction error.

2.3.1. Machine learning targets and required simulation data

To obtain a surrogate model with the lowest mean relative error it is desired to learn from the low-fidelity simulations as well as from the high fidelity simulations simultaneously. A so called multi-fidelity model can be used for this purpose, aiming at the prediction of C_L^{HF} and C_D^{HF} using both VLM and ReFRESCO simulations. The multi-fidelity model should outperform the single-fidelity model trained with only ReFRESCO simulations because it is based on a richer dataset.

2.3.2. Features

Table 2 summarizes which features are used for the forced pitch motion and forced heave motion case. A distinction is made between static (constant) input features such as the geometric parameters and the dynamic (time-varying) input features such as the pitch and heave signal. As described in Nasser et al. (2025), multi-fidelity modeling is achieved by learning a mapping from low- to high-fidelity outputs, without attempting to reconstruct or merge full solution fields. The multi-fidelity models use the low-fidelity force coefficients computed with VLM as additional dynamic input features. Due to the overlapping (nested) design of experiments, these low-fidelity force coefficients are available for each high-fidelity training sample. However, when a prediction is made with the multi-fidelity model one has the choice of doing a cheap low-fidelity VLM simulation to compute the dynamic input features or use a single-fidelity ML model trained with VLM data to approximate the dynamic input features. Since we aim for a fast conceptual design tool, we use a single-fidelity ML model trained with VLM data to obtain the dynamic input features of the multi-fidelity model in this contribution.

2.3.3. Model architecture

Machine learning based prediction of unsteady aerodynamic forces across multiple geometrical shapes was studied extensively in Li et al. (2021). Long Short-Term Memory (LSTM) networks are used to predict unsteady forces on an airfoil, including a changing airfoil geometry. The key to a successful prediction is the ability to combine static and dynamic input features such that the machine learning model can capture temporal patterns and cross-variate information efficiently. This is done by repeating the static features along the time axis and concatenating the result to the dynamic features before they are fed into the LSTM block. Based on the machine learning model proposed in Li et al. (2021), we developed a very similar model, see Fig. 6(a). The alignment stage is exactly the same but a Bi-directional LSTM (Bi-LSTM) is used instead of a uni-directional LSTM. This is particularly useful when the time-series is non-linear and forms a closed loop which is the case for a hysteresis cycle, see Liu et al. (2024). The model architecture can be summarized as follows:

1. **Alignment phase:** Repeat the static features N_t times and concatenate the results with the dynamic input feature (e.g. motion of the foil) to form a single input block.
2. **Bi-LSTM phase:** The Bi-LSTM blocks are repeated N times this means that the full sequence output of a Bi-LSTM layer is used as input for the next Bi-LSTM layer. We used the default hyperbolic tangent (tanh) activation functions for the Bi-LSTM blocks.
3. **Temporal projection phase:** the full output sequence of the last Bi-LSTM layer is connected to a temporal projection block which consists of single layer MLP with linear activation, see for similar architectures (Chen et al., 2023; Liu et al., 2024).

The number of steps to capture a motion cycle is fixed and equal to $N_t = 41$. As a pre-processing step, the data is interpolated to this fixed resolution such that a cycle is always presented consistently to the ML model. In Section 2.3.4 on hyperparameter optimization we vary the number of stacked Bi-LSTM blocks and the number of

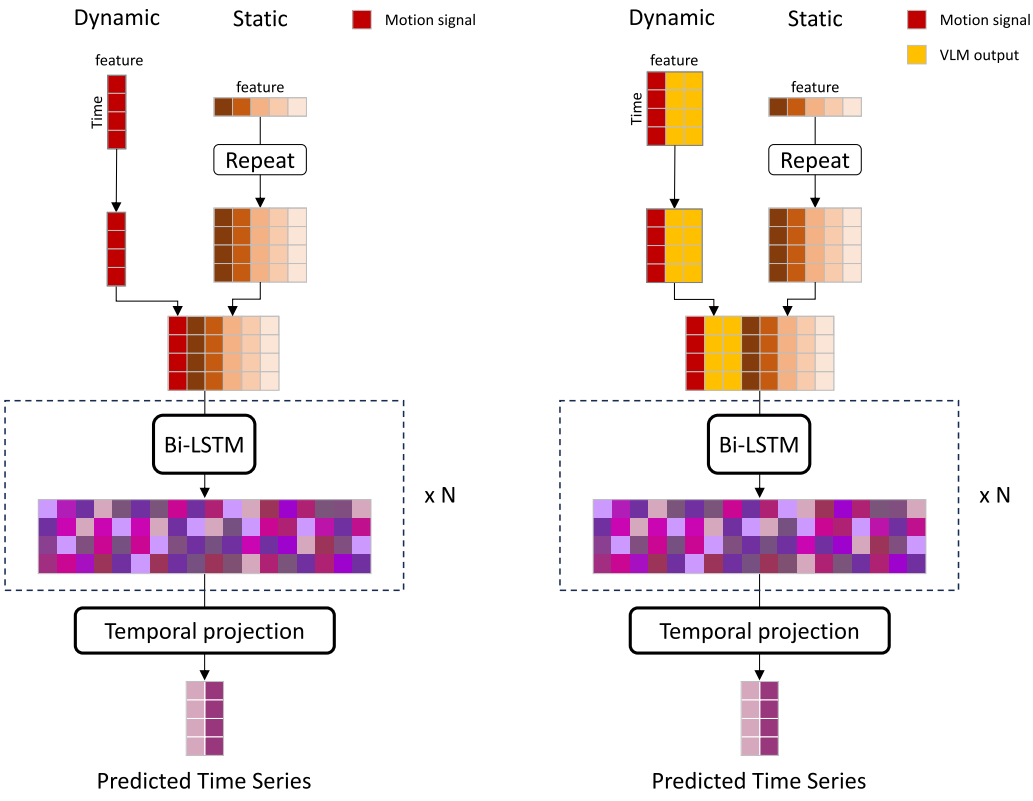


Fig. 6. (a) single-fidelity Bi-LSTM model and (b) multi-fidelity Bi-LSTM model.

Table 2
Dynamic and static features for the single and multi-fidelity models.

	Fidelity	Static features										Dynamic features			
		Fn	Re	Ar	λ_{y_k}	λ_{tap}	λ_{cr}	λ_t	λ_s	k_α	k_h	α	h	C_L^{LF}	C_D^{LF}
Pitch	Single	x	x	x	x	x	x	x	x	x		x			
	Multi	x	x	x	x	x	x	x	x	x		x		x	x
Heave	Single	x	x	x	x	x	x	x	x		x		x		
	Multi	x	x	x	x	x	x	x	x		x		x	x	x

hidden states of the LSTM layers to find the combination that gives optimal performance. Fig. 6(b) shows the multi-fidelity architecture which used the VLM model output as an additional dynamic input feature. This means that the motion signal and low-fidelity VLM output (force coefficients) are stacked and form (together with the repeated static features) a single input block for the Bi-LSTM model. In this way the Bi-LSTM model has a richer input than using the motion signal only.

2.3.4. Hyperparameter optimization and model selection

The training cases used to select the model for deployment in the design tool are defined in Table 3. The first case considers training on data from VLM simulations only. This is an important reference as model performance on VLM data will influence the input of multi-fidelity models, thereby affecting their performance. The second and third case consider single-fidelity LSTM models trained on data from RANS simulations only and the fourth and fifth case consider multi-fidelity LSTM models trained on data from both RANS and VLM simulations. Using this setup we can compare the performance of multi-fidelity models to single-fidelity models and investigate how the ratio of high to low fidelity training data affects model performance.

To facilitate a fair comparison, a hyper-parameter grid search is performed with the learning rate $l_r \in [10^{-3}, 10^{-4}, 10^{-5}]$, the number of Bi-LSTM layers $N \in [1, 2, 4]$, the number of hidden states $M \in [250, 500, 1000]$, the dropout rate $D \in [0, 0.2]$ and the number of epochs $E \in [250, 500, 1000]$ for each case defined in Table 3.

Table 3
Training cases for the foil undergoing pitch and heave motions.

Case	LSTM model	Target	# RANS	# VLM	C_{REF}
1	Single-fidelity	VLM	–	400	VLM
2	Single-fidelity	RANS	50	–	RANS
3	Single-fidelity	RANS	100	–	RANS
4	Multi-fidelity	RANS	50	400	RANS
5	Multi-fidelity	RANS	100	400	RANS

With T_P denoting the period of oscillation and C the prediction of a Machine Learning (ML) model, the Mean Relative Error (MRE) is given by

$$\text{MRE} = \frac{1}{T_p} \int_{\bar{t}=0}^{T_p} \frac{\|C - C_{\text{ref}}\|^2}{\|C_{\text{ref}}\|^2} \mathrm{d}\bar{t} \times 100\% \tag{10}$$

is used to measure the performance of the ML model relative to either the true RANS or VLM reference C_{REF}, see Table 3. A random set of samples with a size of 5 % of the total number of samples is used as a test set on which the test error is computed. The remaining 95 % of

Table 4
Optimal hyperparameters and performance metrics. Bold numbers indicate the lowest validation MRE.

Motion	Target	Case	Lr	Epochs	N	M	Dropout	Test MRE	Val MRE
Pitch	Drag	1	10^{-4}	250	4	500	0.0	1.97 %	4.24 %
		2	10^{-3}	500	4	500	0.0	8.55 %	13.34 %
		3	10^{-3}	250	2	500	0.0	12.44 %	10.12 %
		4	10^{-3}	250	2	500	0.2	6.00 %	10.44 %
		5	$\mathbf{10^{-3}}$	**250**	**2**	**500**	**0.0**	**9.67 %**	**7.46 %**
	Lift	1	10^{-4}	250	4	500	0.0	3.39 %	5.64 %
		2	10^{-3}	250	4	250	0.0	17.10 %	30.52 %
		3	10^{-3}	250	2	500	0.0	18.32 %	19.66 %
		4	10^{-3}	250	4	250	0.0	12.22 %	25.13 %
		5	$\mathbf{10^{-4}}$	**500**	**4**	**250**	**0.0**	**13.16 %**	**17.10 %**
Heave	Drag	1	10^{-3}	1000	4	500	0.0	2.47 %	1.63 %
		2	10^{-4}	250	4	500	0.0	6.18 %	5.72 %
		3	10^{-3}	250	2	500	0.0	3.12 %	6.30 %
		4	$\mathbf{10^{-3}}$	**250**	**2**	**250**	**0.0**	**5.78 %**	**3.66 %**
		5	10^{-3}	250	1	500	0.0	2.20 %	4.29 %
	Lift	1	10^{-3}	250	4	500	0.0	8.72 %	8.72 %
		2	10^{-3}	250	4	500	0.0	15.27 %	41.00 %
		3	10^{-3}	250	4	500	0.0	9.85 %	41.53 %
		4	$\mathbf{10^{-4}}$	**500**	**2**	**500**	**0.2**	**1.78 %**	**33.73 %**
		5	10^{-4}	500	2	250	0.2	7.40 %	34.50 %

the samples are used to define a 5-fold cross-validation set on which the cross-validation error is computed.

Table 4 summarizes the optimal hyperparameters with corresponding performance metrics. The best case is selected on the basis of lowest cross-validation error. Looking at the cross-validation errors of the pitch motion case, we learn that the multi-fidelity model performs best when all high-fidelity data is combined with all low-fidelity data. For the heave motion case, on the other hand, it is found that it works best to include 50 RANS simulations instead of 100 RANS simulations for the training of the multi-fidelity model.

2.3.5. Learning curves

We study the 5-fold mean relative cross-validation error of the single fidelity Bi-LSTM model trained on VLM data (case 1) with increasing number of samples in the design of experiments to see how the prediction of the low-fidelity force coefficients C_L^{LF} and C_D^{LF} improve with the number of samples used for training. Fig. 7 shows the test errors on a set containing 5 % of the total number of samples in each LHS design level and the 5-fold cross-validation errors on a set containing the remaining 95 % of the total number of samples in each LHS design level. The validation error drops with an increasing number of training samples for both the pitch and heave motion case. The performance of the models saturate around N = 400 samples with relative test errors far below 5 % in both lift and drag. This performance is considered sufficient for the low-fidelity prediction model.

2.4. Optimizer

The conceptual design tool should exploit the previously developed surrogate models in order to arrive at a number of initial optimal designs that satisfy all problem constraints. In order to do so, techniques from the field of Simulation Based Design Optimization (SBDO) can be used, see Serani et al. (2024). It should be emphasized that preliminary design is addressed with this tool which includes assumptions on the structure and material of the foil.

2.4.1. Formulation

The time averaged lift and drag coefficient are given by

$$\bar{C}_{L/D}(\mathbf{x}',\mathbf{y}') = \frac{1}{T_p}\int_{t'=0}^{T_p} C_{L/D}(\mathbf{x}',\mathbf{y}',t')dt'. \tag{11}$$

The key objectives in the optimization process are:

1. **Maximize mean lift** (F_L): Lift is a critical objective for hydrofoil design, as it directly impacts the ability of the hydrofoil to support weight by generating a vertical force. Here, a quasi-steady mean lift is used, given by:

$$F_L(\mathbf{x}',\mathbf{y}') = \frac{1}{2}\rho V^2 S\bar{C}_L(\mathbf{x}',\mathbf{y}') \tag{12}$$

2. **Minimize mean drag** ($\bar{F}_D$): Drag reduction is essential for improving foil efficiency, allowing for higher speeds and lower energy consumption. Similar to lift, the quasi-steady mean drag is used as an objective, given by:

$$F_D(\mathbf{x}',\mathbf{y}') = \frac{1}{2}\rho V^2 S\bar{C}_D(\mathbf{x}',\mathbf{y}') \tag{13}$$

3. **Minimize cavitation risk** (σ_{risk}): Cavitation occurs when local pressure drops below vapor pressure leading to bubble formation. It can degrade performance and cause damage to the hydrofoil. Minimizing cavitation risk is therefore essential for ensuring durability and efficiency in foil design. Further details are provided in Section 2.4.2 regarding the description of the cavitation bucket concept and the cavitation objective.

To ensure that the optimized designs are feasible, the following constraints are imposed:

1. **Minimum required lift:** The lift generated by the foil must exceed a specified minimum lift value ($\bar{F}_L^*$) to ensure it can support the necessary operational load.

$$g_L(\mathbf{x}',\mathbf{y}') = F_L^* - F_L(\mathbf{x}',\mathbf{y}') \le 0 \tag{14}$$

2. **Minimum yielding stress:** The foil's structure must maintain its shape and withstand bending stress to ensure structural integrity and manufacturability. This constraint is based on calculating the bending stress for foil sections, which requires the determination of both cross-sectional area (A) and area moment of inertia (I). The area is computed by integrating the difference between the upper and lower surface profiles along the chord:

$$A = \int_0^{c_l} (Z_u - Z_l)\,dx, \tag{15}$$

where c_l is the local chord length. The area moment of inertia is obtained by integrating the vertical distance from the neutral axis (Drela et al., 2024):

$$I = \int_0^{c_l} \frac{(Z_u - \bar{z})^3 - (Z_l - \bar{z})^3}{3}\,dx, \tag{16}$$

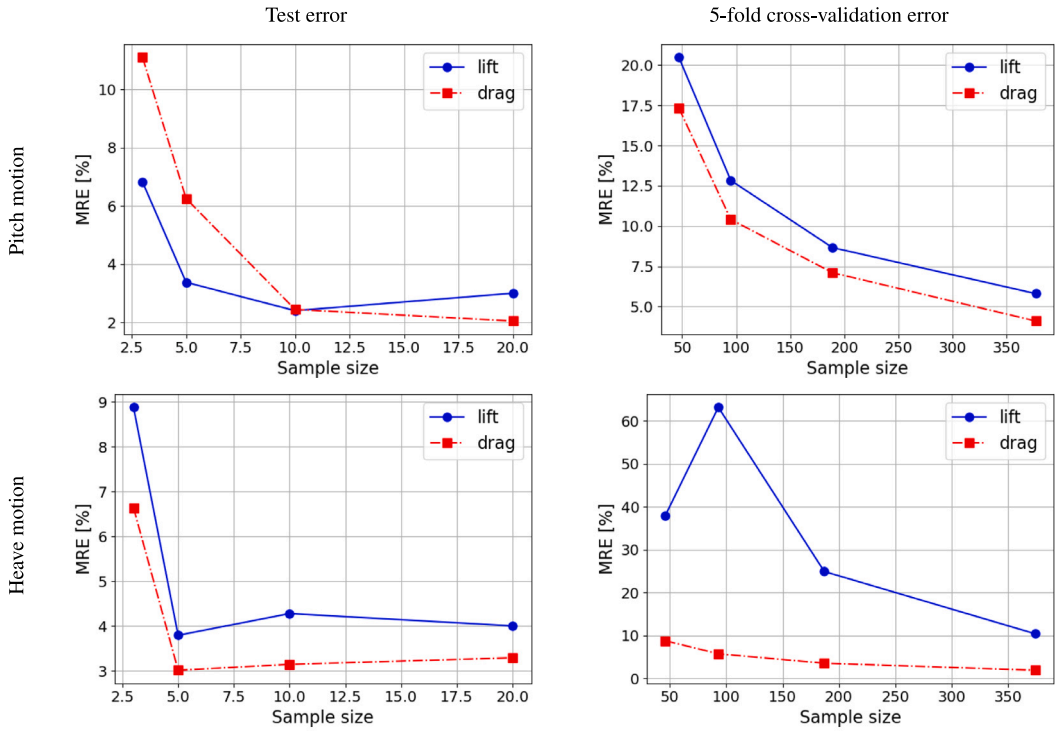


Fig. 7. Learning curves for the pitch and heave motion cases.

where $\bar{z}$ is the neutral axis position. The maximum normal stress due to bending σ_{bend} is calculated from the flexure formula given by

$$\sigma_{\text{bend}} = -\frac{M_{\max} \cdot z_{\max}}{I}, \tag{17}$$

where $M_{\max}$ is the maximum bending moment and $z_{\max}$ is the maximum distance from the neutral axis. The stress is compared to the material's yield stress σ_{yield}. Therefore, the constraint is defined as:

$$g_{\sigma_{bend}} = \max(0, \sigma_{min} - \sigma_b) + \max(0, \sigma_b - \sigma_{max}) \leq 0 \tag{18}$$

where,

$$\sigma_{min} = \sigma_{\text{yield}}/1.30, \quad \sigma_{max} = \sigma_{\text{yield}}/1.15 \tag{19}$$

This constraint ensures that the design remains safe by keeping the bending stress within the yield stress range, with a safety margin between 15 % and 30 %. This range limits the search space, avoiding excessively thick or short foils, which may be structurally sound but economically inefficient due to higher manufacturing costs.

3. **Zero cavitation at neutral foil angle:** The design must avoid cavitation at a neutral angle of attack to ensure reliable performance under typical operating conditions. We define

$$g_{\sigma_{Neutral}} = \sigma_{bucket} - \sigma_{\text{N,neutral}} \leq 0, \tag{20}$$

where σ_{bucket} is the cavitation threshold and σ_{neutral} is the numerically evaluated cavitation number at the neutral angle of attack. Details on cavitation evaluation are provided in Section 2.4.2.

In the context of early-stage hydrofoil exploration and optimization, a series of objectives and constraints are defined to represent the multi-objective minimization problem. For a set of fixed non-dimensional operational input parameters $\mathbf{y}'_1, \mathbf{y}'_2, \ldots, \mathbf{y}'_N$ the following objectives, constraints, and variables are defined;

$$\min_{\mathbf{x}'} \quad \{-F_L(\mathbf{x}', \mathbf{y}'_i),\ F_D(\mathbf{x}', \mathbf{y}'_i),\ -\sigma_{\text{Risk}}(\mathbf{x}')\} \tag{21}$$

$$\begin{aligned} \text{subject to} \quad & g_{\text{L}}(\mathbf{x}', \mathbf{y}'_i) \leq 0, \\ & g_{\sigma_{\text{bend}}}(\mathbf{x}', \mathbf{y}'_i) \leq 0, \\ & g_{\sigma_{\text{N,neutral}}}(\mathbf{x}', \mathbf{y}'_i) \leq 0, \\ \text{given} \quad & \mathbf{x}'_l \leq \mathbf{x}' \leq \mathbf{x}'_u, \quad i = 1, \ldots, N. \end{aligned}$$

When dimensional operational input parameters are fixed, the non-dimensional operational input parameters are written $\mathbf{y}'_1|\mathbf{y}_1, \mathbf{y}'_2|\mathbf{y}_2, \ldots, \mathbf{y}'_N|\mathbf{y}_N$. In that case the minimization in Eq. (21) is taken with respect to $\mathbf{x}'$ and the remaining unconstrained dimensional parameters. The computational cost of the multi-objective optimization is significantly reduced by using surrogate approximations for the objectives and constraints. In that case we solve

$$\min_{\mathbf{x}'} \quad \{-\hat{F}_L(\mathbf{x}', \mathbf{y}'_i),\ \hat{F}_D(\mathbf{x}', \mathbf{y}'_i),\ -\hat{\sigma}_{\text{Risk}}(\mathbf{x}')\} \tag{22}$$

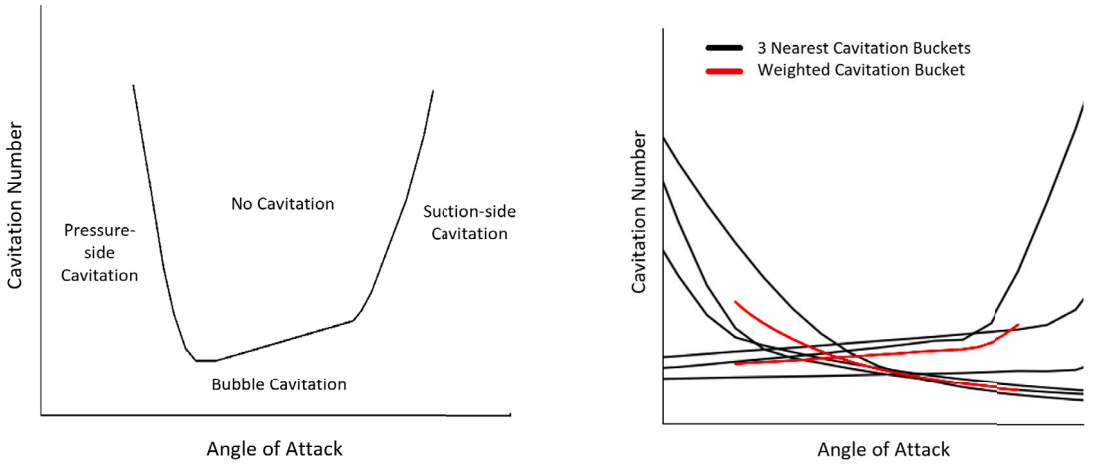
$$\begin{aligned} \text{subject to} \quad & \hat{g}_{\text{L}}(\mathbf{x}', \mathbf{y}'_i) \leq 0, \\ & \hat{g}_{\sigma_{\text{bend}}}(\mathbf{x}', \mathbf{y}'_i) \leq 0, \\ & \hat{g}_{\sigma_{\text{N,neutral}}}(\mathbf{x}', \mathbf{y}'_i) \leq 0, \\ \text{given} \quad & \mathbf{x}'_l \leq \mathbf{x}' \leq \mathbf{x}'_u, \quad i = 1, \ldots, N, \end{aligned}$$

where a hat indicates that surrogate models from Section 2.3 are used to replace the expensive functions. This allows for repeated evaluations across varying thresholds and design boundaries, see Serani et al. (2024).

2.4.2. Cavitation risk quantification

Cavitation buckets for 400 sampled foils were evaluated to quantify cavitation risk accurately. These buckets, computed using PROCAL—a potential flow boundary element method propeller solver, see Bosschers et al. (2008)—describe the cavitation characteristics of the foils for angles of attack between ±15°. Fig. 8 shows a generalized curve highlighting critical cavitation regions.

For small angles of attack, the inception curves are relatively flat, suggesting that operating at low cavitation numbers will typically lead to bubble-type cavitation, which usually occurs mid-chord on the blade. In contrast, at large angles of attack, the steep inception curves suggest



(a) Generalized cavitation bucket with critical cavitating regions

(b) Example of a 3 Nearest neighbour weighted cavitation bucket

Fig. 8. Cavitation bucket description using K-NN methodology.

a pressure peak at the leading edge. Operation beyond this curve (at low cavitation numbers and high angles of attack) would result in sheet cavitation on the foil. In design optimization, new foil geometries have unique cavitation buckets that must be re-evaluated, presenting a challenge. To address this, we used a nearest-neighbor approach with existing Latin Hypercube Sampling (LHS) points. The algorithm leverages costly pre-calculated simulations by adopting the k-nearest geometrically similar foils from an existing database. This approach enables quick and accurate approximations of cavitation behavior for new foil designs. A weighted 3-nearest neighbor using a euclidean distance (3 geometrically closest foils) algorithm was implemented to maintain the optimized foil's unique properties. From these newly evaluated buckets, the risk of cavitation can be quantified using a continuously defined objective function:

$$\sigma_{risk} = \frac{\sum(RF \cdot \Delta)}{\sum RF}, \tag{23}$$

where Δ is given by

$$\Delta = \max(\sigma_{\mathrm{N,pressure}}, \sigma_{\mathrm{N,suction}}) - \sigma_{\mathrm{N,neutral}}. \tag{24}$$

The objective function quantifies cavitation risk by calculating the sum of differences between the foil's cavitation number ($\sigma_{\mathrm{N,\ neutral}}$) and the bucket boundaries across a range of angles of attacks on both the suction and pressure sides ($\sigma_{N,suction}, \sigma_{N,pressure}$). A lower value of this objective function indicates a higher risk of cavitation. Cavitation risk varies depending on the type of cavitation present. Thus, the Risk Factors (RF) for different cavitation types are defined as follows:

- Suction side sheet cavitation (less severe): RF = 1
- Pressure side sheet cavitation (more severe): RF = 2
- Bubble cavitation (most severe): RF = 3

It should be noted that the corresponding relation holds no physical unit representation and as such should be observed purely from a relative sense.

2.4.3. Algorithm

Genetic algorithms (GAs) are nature-inspired optimization methods that evolve a population of candidate solutions through selection, crossover, and mutation. The NSGA-III (Non-dominated Sorting Genetic Algorithm III) (Blank and Deb, 2020), was employed in this study for early-stage foil design optimization. NSGA-III is particularly suited to this task due to its ability to balance multiple objectives and handle non-linear, multi-modal problems effectively. Its global search capability enables a comprehensive exploration of the design space, though GAs can be computationally expensive and sensitive to parameter tuning (e.g., population size and mutation rate). Key parameters of the algorithm are empirically tuned to balance computational cost with solution quality:

1. **Population size:** A size of 100 is selected to ensure sufficient diversity while maintaining computational efficiency.
2. **Maximum generations:** The optimization is capped at 10 generations. This number is based on preliminary convergence tests indicating that this range ensures feasible computation time without sacrificing solution quality.
3. **Mutation rate:** A default mutation rate of 0.15 % adopted based on [Y] testing, ensuring diversity without disrupting population stability.

In addition to these hyperparameters, NSGA-III uses specialized reference points to promote diversity, ensuring solutions are well-distributed along the Pareto front. This enables a wide range of trade-offs between lift, drag, and cavitation risk, giving decision-makers flexibility in selecting optimal designs. The Pareto front represents the set of non-dominated solutions, where improving one objective (e.g., lift) would worsen another (e.g., drag). To select the final foil design, a weighted sum of normalized objectives is used, allowing prioritization based on user-defined criteria:

$$\text{Weighted Objective} = w_1 \cdot \bar{F}_L + w_2 \cdot \bar{F}_D + w_3 \cdot \sigma_{risk}, \tag{25}$$

where w_1, w_2, w_3 represent the relative importance of each objective. Adjusting these weights tailors the design to meet specific performance requirements.

3. Conceptual design tool development

Implementing the functionalities of the conceptual design tool as indicated in Fig. 1 is a challenge. In addition to a correct functional implementation, the tool should be fast, intuitive and maintainable over a long period of time.

3.1. Functionality and requirements

A web application has been developed with three requirements in mind: accessibility, updateability of the models and scalability of computations. The genetic algorithm described in Section 2.4.3 has been encapsulated within a user-friendly interface to facilitate interactive exploration and evaluation of foil designs. The application offers two modes of operation: optimization and evaluation. In optimization

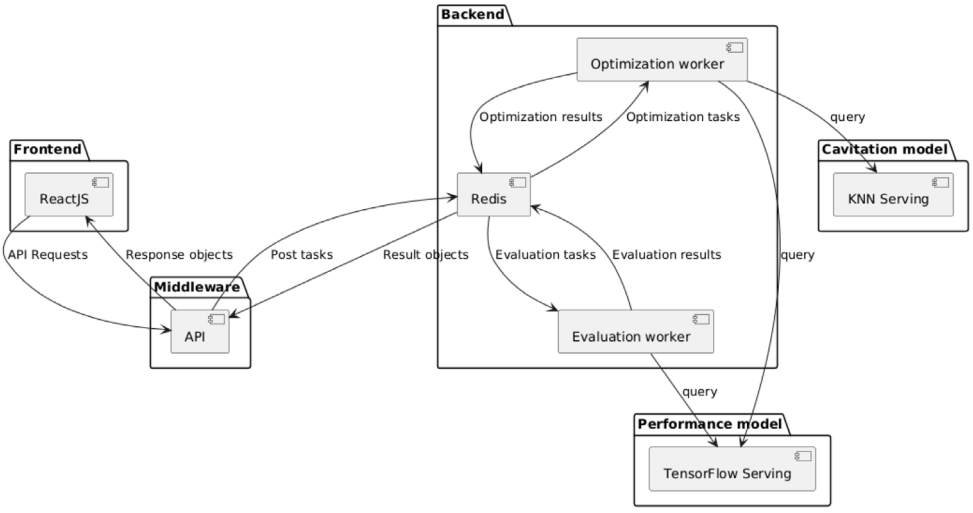


Fig. 9. Schematic of the tool architecture.

mode, the genetic algorithm executes iterative optimization within constraints set by the user, see Section 2.4.1. All results that satisfy the constraints are presented to the user, including evaluation results for the foil design that best fits the design requirements. In evaluation mode, the user can submit foil characteristics and operational settings to study the pitch and heave behavior of the foil.

3.2. Application setup and graphical user interface

The application has been set up as a collection of Docker containers, see Fig. 9. Tensorflow models are wrapped in a TF_Serving container (TensorFlow, 2024) allowing them to be queried over HTTP through a REST API, see Fielding and Taylor (2000). Similarly, the K-NN model for cavitation evaluation described in Section 2.4.2 has been made accessible through a Flask API in a separate docker container, see Grinberg (2018). The models can be updated or replaced without interrupting the availability of the front-end or back-end because they are deployed in containers independent of the front-end and the back-end. To address the computational intensity of the genetic algorithm, which may take several minutes per run, the API has been implemented asynchronously. This ensures that the front-end remains responsive while computations are running in the background. Each optimization and evaluation request is posted on a task queue. Workers in different containers can read from the queue and execute the tasks. The number of workers can be scaled up or down based on user demand, ensuring efficient resource utilization and performance under varying loads.

An impression of the graphical user interface is given in Fig. 10. The application starts in the optimization mode where the user can specify operational conditions and design parameter ranges for the optimization process. After completion of the optimization, the Pareto front is presented to the user in the graphical user interface, see Fig. 11. The user can click on points on the Pareto front and directly see the corresponding time-dependent graphs. By using the tabs the designer can easily switch from optimization to evaluation mode and vice-versa. The evaluation mode can be used for an in-depth analysis of the designs.

4. Verification and validation

To perform Verification and Validation (V&V), the outcome of the design tool is compared to RANS and experimental results respectively. The experimental results were obtained by model tests performed by Delft University of Technology, see Fig. 12. More information on the model test campaign and experimental setup can be found in Minerva et al. (2024). The hydrofoil we select for V&V has an aspect ratio $A_r = 8$, a kink distance ratio $\lambda_{y_k} = 0.73$, a tapering ratio $\frac{c_{tip}}{c_{root}} = 0.554$ and a submergence ratio of $\lambda_s = 0.14$. Furthermore, a foil profile based on Eppler 817 is used with camber ratio $\lambda_{cr} = 0.029$ and thickness ratio $\lambda_t = 0.2$. The hydrofoil is pitching while having forward speed, resulting in a Froude number $F_n = 4.14$ and a Reynolds number $Re = 3.455 \cdot 10^5$. The Reynolds number is unfortunately not within the limits of the design of experiments used for training the machine learning model since $3.455 \cdot 10^5 < 1.01 \cdot 10^6$. The results should therefore be interpreted carefully when making the comparison to RANS and the experiments. The prediction error of the conceptual design tool consists of an error due to geometric approximation of the hydrofoil and a machine learning error, which in this case is affected by the fact that one feature is out of bounds. For the machine learning model we used the multi-fidelity model trained with all low and high-fidelity data (case 5 from Section 2.3.4).

The verification and validation of the lift coefficients due to pitching of the hydrofoil are shown in Fig. 13. Two different cases are considered: A quasi-static case with a pitching hydrofoil at reduced frequency $k = 0.006$ and amplitude $a_\alpha = 10$ degrees, shown in Fig. 13(a), and a dynamic case at reduced frequency $k = 0.131$ and amplitude $a_\alpha = 3.5$ degrees, shown in Fig. 13(b). The reduced frequency values are very close but not identical to the values used in Simonet et al. (2024) ($k = 0.008$ and $k = 0.14$) to obtain a quasi-steady and a dynamic flow case. In both cases it is verified that the machine learning prediction closely follows the high-fidelity RANS results. For the higher reduced frequency case, the hysteresis loop is however narrower than the results from RANS indicate. This could be the result of the fact that the Reynolds number is out of feature bounds since Reynolds can have an effect on not only drag but also on lift and thereby on the shed vorticity which in turn affects the width of the hysteresis loop of the pitching hydrofoil. The measured lift coefficient is systematically lower than the predictions of RANS and therefore also lower than the predictions of the machine learning model. A plausible explanation for the reduced experimental performance is the effect of the vertical strut which is not modeled in the RANS computations:

1. It was shown that for low Reynolds numbers the lift of a T-shaped foil can be reduced significantly by the vertical strut, see AlaviMehr et al. (2015). Here, the lift curve slope in the steady case was found to be 61 % of the value of an ideal lift slope for a foil with elliptic loading.
2. Differences in viscosity effects such as boundary layer transition, which is not properly modeled in RANS, may also contribute

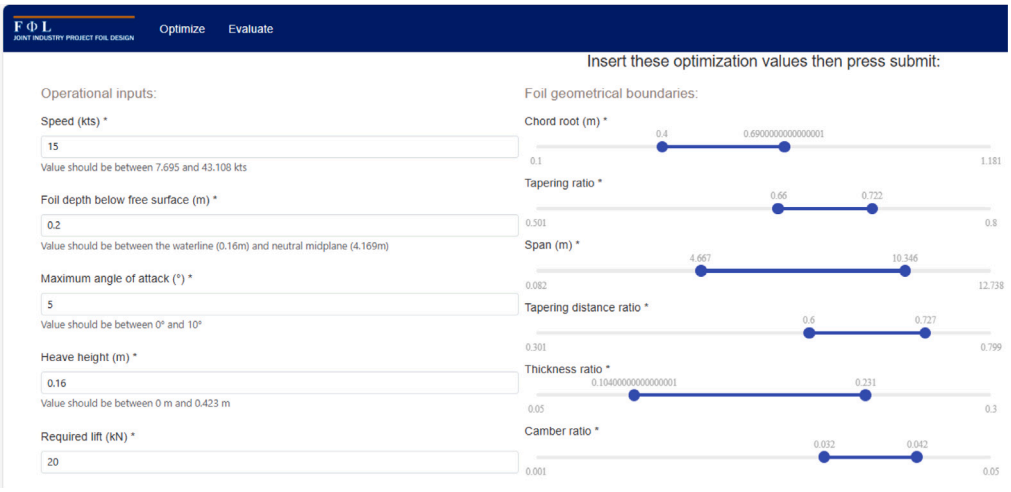


Fig. 10. Impression of the graphical user interface: user input for optimization.

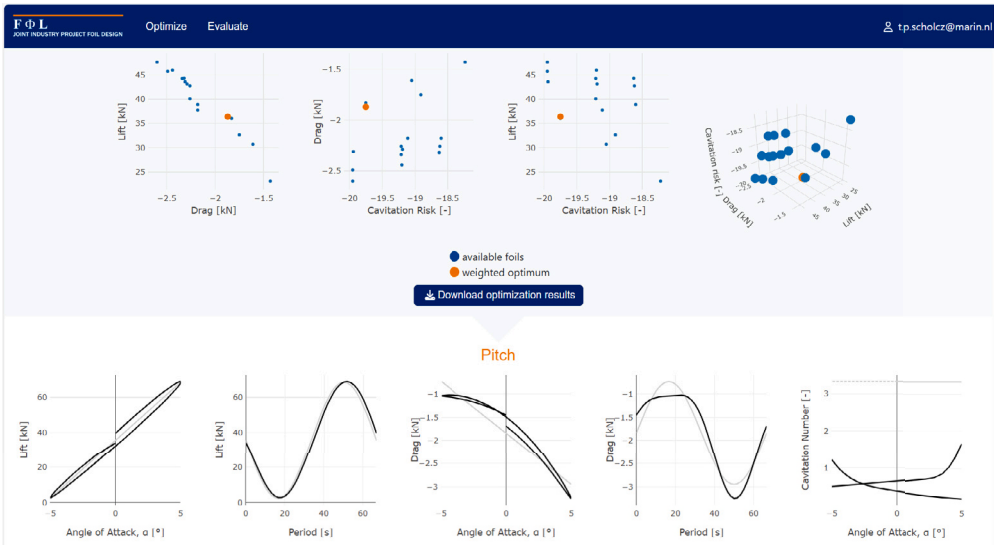
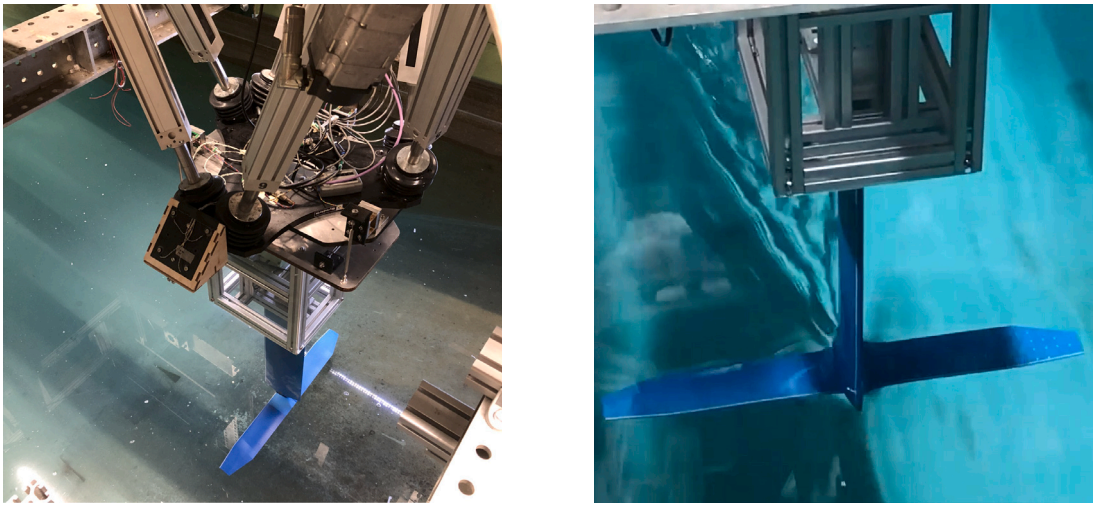


Fig. 11. Impression of the graphical user interface: optimization output presented to the user.



(a) Experimental setup of the measurement campaign.

(b) Snapshot during an experimental run.

Fig. 12. Footage of hydrofoil model tests performed by Delft University of Technology.

Table 5
Dimensional features of the optimized hydrofoils with increasing lift targets.

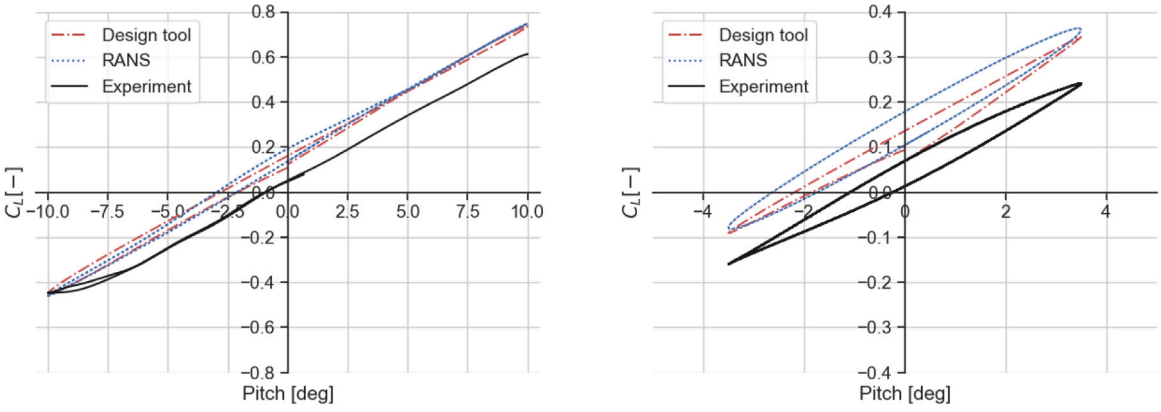
Target (±10 N)	Weights	Geometrical (**x**)							Operational (**y**)		Tool		RANS	
		$\text{cam}_{\max}$	c	c_{tip}	$t_{\max}$	y_k[a]	S[a]	b[a]	U_∞	d	F_L	F_D	F_L	F_D
Lift = 1000 N	Equal	0.021	0.144	0.107	0.024	0.684	0.167	1.158	10.0	0.16	1006	123	1079	126
Lift = 1000 N	Lift only	0.003	0.153	0.122	0.030	0.557	0.400	2.620	10.0	0.16	**1009**	310	–	–
Lift = 1000 N	Drag only	0.021	0.144	0.111	0.011	0.547	0.121	0.838	10.0	0.16	999	**81**	–	–
Lift = 2000 N	Equal	0.006	0.158	0.088	0.021	0.704	0.140	0.884	10.0	0.16	2008	146	1683	124
Lift = 2000 N	Lift only	0.003	0.121	0.094	0.022	0.900	0.297	2.451	10.0	0.16	**2009**	211	–	–
Lift = 2000 N	Drag only	0.006	0.159	0.006	0.021	0.686	0.140	0.880	10.0	0.16	1993	**145**	–	–
Lift = 4000 N	Equal	0.004	0.156	0.099	0.037	1.611	0.833	5.346	10.0	0.16	4001	564	3757	630
Lift = 4000 N	Lift only	0.004	0.158	0.111	0.037	1.580	0.828	5.245	10.0	0.16	**4010**	598	–	–
Lift = 4000 N	Drag only	0.005	0.156	0.095	0.032	1.710	0.588	3.773	10.0	0.16	3995	**410**	–	–

[a] We report the values for the full hydrofoil here for comparison.

Table 6
Non-dimensional features of the optimized hydrofoils with increasing lift targets.

Target (±10 N)	Optimization weights	Geometrical (**x′**)					Operational (**y′**|**y**)			Output		
		Ar[a]	λ_{y_k}	λ_{tap}	λ_{cr}	λ_t	Fn	Re	λ_s	$\bar{C}_L$	$\bar{C}_D$	σ_{risk}
Lift = 1000 N	Equal	8.054	0.591	0.695	0.147	0.167	8.414	$1.309 \cdot 10^6$	1.111	0.121	0.0147	11.923
Lift = 1000 N	Lift only	17.087	0.213	0.758	0.0169	0.196	8.162	$1.391 \cdot 10^6$	1.046	0.0506	0.0155	10.140
Lift = 1000 N	Drag only	5.835	0.653	0.736	0.143	0.0764	8.414	$1.309 \cdot 10^6$	1.111	0.165	0.0134	12.842
Lift = 2000 N	Equal	5.608	0.796	0.530	0.038	0.133	8.032	$1.436 \cdot 10^6$	1.013	0.287	0.0210	11.293
Lift = 2000 N	Lift only	20.285	0.367	0.701	0.025	0.182	9.179	$1.100 \cdot 10^6$	1.322	0.136	0.014	9.013
Lift = 2000 N	Drag only	5.538	0.780	0.036	0.038	0.132	8.007	$1.445 \cdot 10^6$	1.006	0.285	0.021	11.297
Lift = 4000 N	Equal	34.229	0.301	0.532	0.026	0.237	8.084	$1.418 \cdot 10^6$	1.026	0.096	0.014	10.618
Lift = 4000 N	Lift only	33.168	0.301	0.607	0.025	0.234	8.032	$1.436 \cdot 10^6$	1.013	0.097	0.014	9.013
Lift = 4000 N	Drag only	24.162	0.453	0.531	0.032	0.205	8.084	$1.418 \cdot 10^6$	1.026	0.136	0.014	11.297

[a] We report the values for the full hydrofoil here for comparison.



(a) Lift coefficients for $k = 0.006$ and $a_\alpha = 10$ degrees.

(b) Lift coefficients for $k = 0.131$ and $a_\alpha = 3.5$ degrees.

Fig. 13. Verification and validation results for the pitching hydrofoil case: lift coefficients.

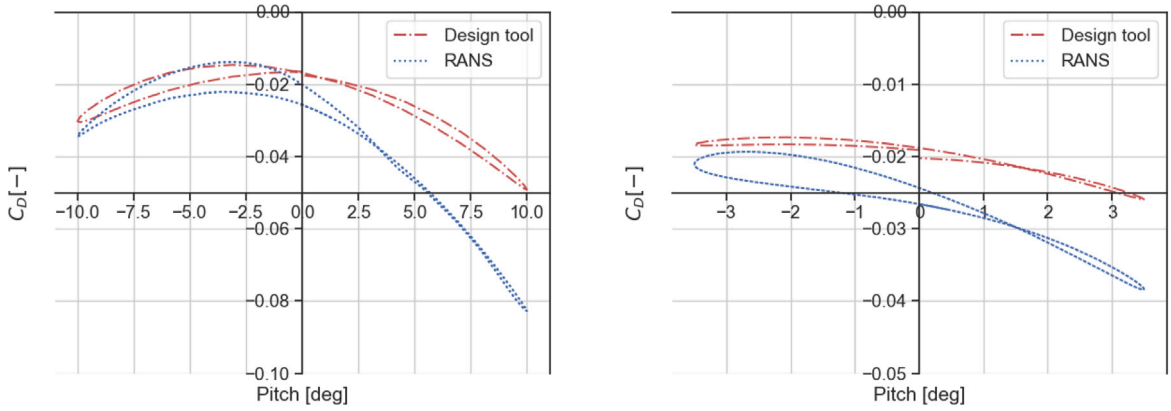
to the bias. Turbulent flow during the experiments may lower the measured lift compared to the RANS calculations where transition did not take place.

Fig. 14 shows the verification results of the drag coefficients due to pitching of the hydrofoil. It can be seen that the machine learning model predicts a narrower hysteresis loop when compared to the RANS results, especially for reduced frequency $k = 0.131$. The prediction accuracy is however still much better than the accuracy that we would obtain when we relied on VLM only. We conclude that for the quasi-static and dynamic case the lift and drag prediction of the design tool was verified to be close to the RANS results, with a somewhat narrower hysteresis loop for the dynamic case. The validation results show a bias in lift coefficient which are attributed to geometry differences and RANS modeling errors. Overall, we observe sufficient performance of the design tool for the conceptual design of hydrofoils.

5. Case study

We will now use the conceptual design tool to optimize pitching foil geometries with $k = 0.012$ and $a_\alpha = 3.5$ while constraining the dimensional speed and submergence to $U_\infty = 10.0$ meters per second and $d = 0.16$ m respectively, see Table 5. Three targets are defined for the dimensional lift: 1000, 2000 and 4000 N. The selected speed and targets are representative of small craft at high speed. We allow the lift to vary in a range of ±10 N with respect to the target value and for each target we present the optimization results with the objectives weighted in the following way:

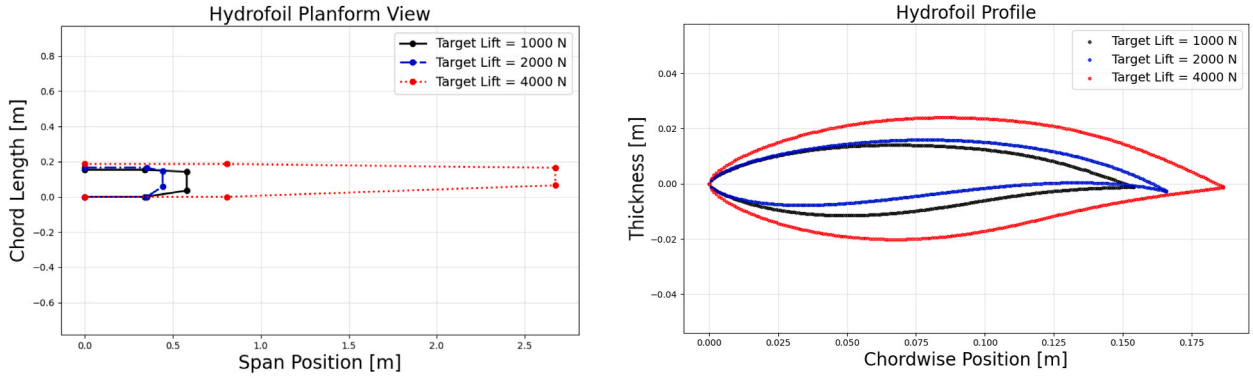
1. Equally: $w = (1/3, 1/3, 1/3)$ on lift, drag and cavitation.
2. Lift only: $w = (1, 0, 0)$ on lift, drag and cavitation.
3. Drag only: $w = (0, 1, 0)$ on lift, drag and cavitation.



(a) Drag coefficients for $k = 0.006$ and $a_\alpha = 10$ degrees.

(b) Drag coefficients for $k = 0.131$ and $a_\alpha = 3.5$ degrees.

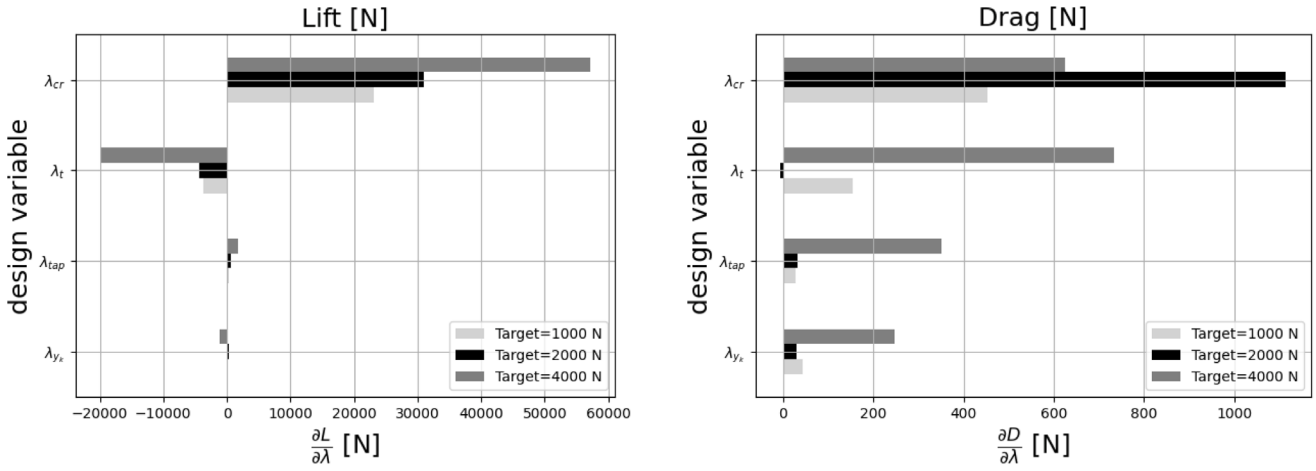
Fig. 14. Verification results for the pitching hydrofoil case: drag coefficients.



(a) Planform designs.

(b) Hydrofoil profiles

Fig. 15. Plot of the optimized hydrofoil geometries corresponding to Tables 5 and 6 with equal optimization weights.



(a) Lift.

(b) Drag.

Fig. 16. Sensitivity gradient analysis of the optimized hydrofoil geometries corresponding to Tables 5 and 6 with equal optimization weights.

For each case we observe that the optimization for pure lift and drag indeed results in the highest lift and lowest drag (displayed in bold), as expected. The optimization result with equal weights is a compromise with lower lift, higher drag and lower cavitation objective than would be possible by optimizing on these objectives individually. To verify the tool predictions on lift and drag for these geometries we use RANS predictions, see the last columns of Table 5. Most of the relative errors on lift and drag are below 15 %, which is what we would expect from Table 4 (test error 13 % on lift and 10 % on drag). The optimized geometries with equal optimization weights are shown in Fig. 15.

As can be seen, the geometries are diverse in planform but also in foil profile. Note that the non-dimensional operational features for constrained dimensional lift and submergence, denoted by $\mathbf{y}'|\mathbf{y}$, vary as a consequence of geometry changes, see Table 6. To gain more insight in the optimization results we perform a gradient sensitivity study of the dimensional lift and drag with respect to the non-dimensional design parameters λ_{y_k}, λ_{tap}, λ_{cr} and λ_{t}, see Fig. 16. The sensitivities are computed using finite difference approximations. The most important parameter for lift and drag is the camber ratio of the profile, followed by the thickness ratio. The tapering ratio and kink distance ratio have a smaller but nevertheless important effect on the lift and drag. To demonstrate how the sensitivities could be used consider the "lift only" case with a target of 4000 N in Table 6. This design is quite similar to the "weights equal" case except for the non-dimensional parameters λ_{tap}, λ_{t} and λ_{cr}. Using a Taylor series approximation around the "weights equal" case for the lift we find

$$\begin{aligned}\tilde{L}^{\text{new}} = L^{\text{old}} &+ \frac{\partial L}{\partial \lambda_{\text{tap}}}(\lambda_{\text{tap}}^{\text{new}} - \lambda_{\text{tap}}^{\text{old}}) + \\ &\frac{\partial L}{\partial \lambda_{\text{t}}}(\lambda_{\text{t}}^{\text{new}} - \lambda_{\text{t}}^{\text{old}}) + \\ &\frac{\partial L}{\partial \lambda_{\text{cr}}}(\lambda_{\text{cr}}^{\text{new}} - \lambda_{\text{cr}}^{\text{old}}).\end{aligned} \tag{26}$$

Using this Taylor series we find an estimate for the lift of the "lift only" case of $\tilde{L}^{\text{new}} = 4001 + 1737 \cdot (0.607 - 0.532) - 19981 \cdot (0.234 - 0.237) + 57158 \cdot (0.025 - 0.026) = 4134$ N. The lift decreases due to a reduction in camber but increases due to an increase in tapering ratio and a reduction in thickness ratio. The net result is an increase in lift. Similarly for the drag we find $\tilde{D}^{\text{new}} = 564 + 350 \cdot (0.607 - 0.532) + 733 \cdot (0.234 - 0.237) + 624 \cdot (0.025 - 0.026) = 587$ N. We observe that the increasing tapering ratio is mainly responsible for the increasing (induced) drag. The actual lift and drag values of the "lift only" case are 4010 N and 598 N respectively which shows that the sensitivities can be used to find the sign of lift and drag change due to small geometrical changes but should not be used to obtain accurate estimates. From the case study we conclude that the conceptual design tool quickly yields hydrofoil geometries that satisfy the required performances with acceptable accuracy.

6. Conclusions

Designing a hydrofoil is a challenge since the hydrodynamics is governed by complex physical phenomena such as the effect of the free surface and unsteadiness due to waves and/or operations. Since high-fidelity simulations are often too expensive there is a need for conceptual design tools that quickly yield optimized hydrofoil designs for varying optimization objectives and constraints. At the core of such a tool lies shape parametrization, high and low-fidelity numerical solvers, surrogate models and a multi-objective optimizer. We successfully train a multi-fidelity Bi-LSTM model on a training set involving unsteady low and high-fidelity forced pitch or heave simulations. Moreover, static features that determine the geometry and dynamic features describing the motion of the hydrofoil are combined in the machine-learning model such that the unsteady response of a hydrofoil due to forced pitch and heave motions is predicted along with the geometry of the hydrofoil. After hyper-parameter optimization, the mean relative test error in lift and drag due to forced pitching is 13.16 % and 9.67 % respectively. Similarly, for forced heave, we find 1.78 % and 5.78 % respectively. These percentages show the capability of the conceptual design tool to quickly give a fair estimate of the hydrodynamical performance when the static and dynamic features are within the bounds used for training. The optimization routines were implemented in a web application which was developed with accessibility, updateability of the models and scalability of the computations in mind. Verification and validation was performed by comparing with results from RANS computations and experiments. The predictions of the tool are close to the RANS results but have a more narrow hysteresis loop. For the validation, a bias is observed in the lift which is likely due to the missing strut in the simulations. Using the conceptual design tool, hydrofoils are optimized with increasing lift targets while keeping the dimensional operational conditions fixed. The results demonstrate the practical use of the tool by quickly proposing several designs with favorable performances. This motivates to explore new machine learning models and features in the tool development with the purpose to enhance conceptual hydrofoil design.

CRediT authorship contribution statement

T.P. Scholcz: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **K. Odendaal:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **G. Marelli:** Visualization, Software, Resources, Investigation, Formal analysis. **L. Minerva:** Supervision, Project administration, Methodology, Conceptualization. **O.V. Schnitzeler:** Visualization, Software, Conceptualization. **F. van Walree:** Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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